Question	Answer	Marks	Partial Marks
14(a)	2, 0, 12	**3**	**B1** for each correct value
14(b)	Correct curve	**4**	**B3FT** for 7 or 8 correct points or **B2FT** for 5 or 6 correct points or **B1FT** for 3 or 4 correct points
14(c)	$y = 10 - x$ ruled	**M2**	**B1** for [y =] $10 - x$ soi or $y = k - x$ ruled or $y = 10 + kx$ ruled, but not $y = 10$
	$x = -1.7$ to -1.4 $x = 2$ $x = 4.4$ to 4.7	**A2**	**A1** for any two values If 0 scored, **SC1** for 3 correct values
15(a)	192 or 192.4… nfww	**4**	**M1** for midpoints soi (170, 182.5, 187.5, 195, 205) **M1** for use of Σfm with m in correct interval including both boundaries **M1** (dep on second M1) for $\Sigma fm \div 140$
15(b)	$\frac{69}{278}$ oe	**2**	**M1** for $\frac{70}{140} \times \frac{69}{139}$
15(c)	Correct histogram	**3**	**B2** for three correct blocks or **B1** for two correct blocks If 0 scored, **SC1** for three of $\frac{7}{20}, \frac{12}{5}, \frac{31}{5}, \frac{70}{10}$ or 0.35, 2.4, 6.2, 7
16	2.9[0] or 2.901…	**3**	**M2** for $\sqrt[6]{\frac{1484}{1250}}$ or **M1** for $1250 \times k^6 = 1484$
17(a)	10.4 or 10.42…	**3**	**M2** for $\sqrt{9.6^2 + 12.1^2 - 2 \times 9.6 \times 12.1 \times \cos 56}$ OR **M1** for $9.6^2 + 12.1^2 - 2 \times 9.6 \times 12.1 \times \cos 56$ **A1** for 108.6 to 108.7

Question	Answer	Marks	Partial Marks
17(b)	107.2 to 107.3	**4**	**B3** for 72.7[4…] OR **M3** for $180-\sin^{-1}\left(15.8\times\frac{\sin 47}{12.1}\right)$ **M2** for [sinBDC =] $15.8\times\frac{\sin 47}{12.1}$ or **M1** for $\frac{\sin BDC}{15.8}=\frac{\sin 47}{12.1}$
17(c)	89.59 to 89.80…	**3**	**M1** for $\frac{1}{2}$ × 9.6 × 12.1 × sin56 **M1** for $\frac{1}{2}$ × 15.8 × 12.1 × sin(133 – *their* $\angle BDC$)
18	a = 2 b = 3 c = 20	**3**	**B2** for any two correct For c, FT 2 + *their* a × *their* $(b)^2$ or **B1** for $b=3$ nfww OR **B2** for $2(x+3)^2-20$ or **M1** for $(x+3)^2$ seen or for $ax^2+2abx+ab^2-c$ or for $a=2$ and one of $2ab=12$ and $ab^2-c=-2$
19	45.6 or 45.63…	**4**	**B3** for 30.6 or 30.63… OR **M3** for $\frac{360-126}{360}$ × 2 × π × 7.5 + 15 or **M2** for $\frac{360-126}{360}$ × 2 × π × 7.5 or **M1** for $\frac{126}{360}$ × 2 × π × 7.5 or **B1** for major sector angle = 234